GenCourseAI

Building REST APIs with Node & Express

Learn the fundamentals of backend APIs: routing, parsing request parameters, request validation, middleware flows, and controller logic.

CURRICULUM COMPILED DYNAMICALLY BY **GENCOURSE AI** ENGINE

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MODULE 1

Module 1: Express Routing & Requests

1.1 Understanding REST Constraints

LESSON OBJECTIVES

- Understand the core concepts of 1.1 Understanding REST Constraints
- Learn the best practices of 1.1 Understanding REST Constraints
- Apply 1.1 Understanding REST Constraints in real-world scenarios

Representational State Transfer (REST) is an architectural style for designing networked applications. It relies on a stateless, client-server cacheable communication protocol, typically HTTP.

Key REST Constraints:

- Client-server separation.
- Stateless requests: Each request contains all information needed.
- Uniform Interface: Standard HTTP methods (GET, POST, PUT, DELETE).

1.2 Writing Express Routes & Middlewares

LESSON OBJECTIVES

- Understand the core concepts of 1.2 Writing Express Routes & Middlewares
- Learn the best practices of 1.2 Writing Express Routes & Middlewares
- Apply 1.2 Writing Express Routes & Middlewares in real-world scenarios

Routing determines how an application responds to a client request to a particular endpoint. Middlewares are functions that have access to the request object, response object, and the next function.

```
const checkAuth = (req, res, next) => {
  if (!req.headers.authorization) {
    return res.status(401).json({ error: "Unauthorized" });
  }
  next();
};

router.get("/data", checkAuth, getData);
```

Middlewares intercept requests to run validations or logger updates.